Research on Efficient Landslide Prediction Approaches using Machine Learning Techniques

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***Abstract*—A landslide is a condition in which a huge amount of rock particles slide or break off down a slope, resulting in great natural and physical loss in addition to the lives of many people. In large parts of the world, massive damage is caused by landslides. The utility of remotely sensed images is used for landslide detection, mapping, prediction, and assessment round the world. This systematic analysis might also make contributions to better expertise the considerable use of remotely sensed records and spatial evaluation techniques to conduct landslide research at more than a few scales. The machine learning algorithms in particular ANN and SVM are used as soft computing techniques for landslide prediction. The accuracy obtained from SVM is 91.78% and with ANN 93.38%. In India landslide is famous phenomena of Himalayan location, Western Ghats and southern Nilgiris Mountains. Such losses must be avoided if right perception tool is available that would notify about the event in boost. With the use of proposed soft computing techniques this paper projects unique landslide prediction techniques with cognizance on western India.**

Keywords—Landslide Prediction, Detection, Remote Sensing, GIS, Satellite Images, Machine Learning.

I. Introduction

Landslide is one of the most expensive and deadly geological risks, threatening and influencing the socioeconomic situations in many nations globally [1]. Remote sensing processes are extensively utilized in landslide research. Landslide threats can also be investigated via slope balance model, susceptibility mapping, threat assessment, hazard evaluation, and other strategies. Even though it is viable to do landslide studies the usage of in-situ observation, it's time gaining, expensive, and from time to time hard to collect records at inaccessible terrains. Remotely sensed records may be utilized in landslide monitoring, mapping, threat prediction and assessment, and other investigations [2]. The main aim of this research is to study the existing remote sensing strategies and strategies used to examine landslides and explore the opportunities of capacity remote sensing tools that could correctly be utilized in landslide prediction using SVM & ANN in the future. This chapter additionally presents important and complete evaluations of landslide studies focusing on the function played by means of remote sensing information and soft computing strategies in landslide prediction assessment. Similarly, the reviews projects about the application of remotely sensed images for landslide detection, mapping, prediction, and assessment around the arena [3]. This systematic assessment may make contributions to better knowledge the appropriate use of remotely sensed records and spatial analysis strategies to monitor landslide research at more than a few scales [4].

The remote detecting images applied under many examinations for the landslide forecast or identity or challenge utilizing the distinctive image managing activities & AI techniques [5]. The change identification dependent on the pre & post landslide images carried out making use of exceptional techniques, for example, thresholding, parallel fleeting techniques, surface examination & so on although, all such arrangements gone through massive difficulties identified alongside adaptability, precision, & effectiveness. This raises us to introduce the systematic research of all such techniques provided at some stage in most current twenty years & examine them rather as indicated through the above problems. This paper provides the efficient use of SVM & ANN machine learning algorithms for the landslide prediction. Phase II, affords the assessment of such strategies. Phase III, gives the landslide prediction approaches SVM & ANN. Phase IV, and projects the results of mentioned machine learning algorithms. Phase V-VI presents the result and conclusion.

II. Literature Review

Evaluation Assessment of uncertainties related to landslide prediction is crucial for enhancing the reliability of landslide early warning systems. Landslide prediction using remote sensing imagery has received considerable attention considering that real-time imagery has been used for the past decade; but it suffers from numerous challenging situations like the loss of satisfactory records along with landslide areas data especially in Western and Northern India, irrelevant approach for landslide remote sensing photo research, poor accuracy of landslide analysis, and challenging methods of landslide reforms, localization, and so on [4-5].

Technological advances have provided kind of equipment which if properly assembled can deliver excellent estimation approach. Every approach developed may have its own advantages and drawbacks which method to use depends in large part on place below consideration because of massive diversifications and combinations of triggering elements. Recent technologies or information base control systems and facts representation are easily available these days. Remote sensing technology uses lesser power; greater range and lesser bandwidth are very beneficial.

The technologies of remote sensing and GIS have a special benefit for the sensitivity, evaluation of landslide areas, prediction and localization. The GIS parameters like NDVI, humidity, temperature and so forth are very much useful for the landslide prediction.

In [6], the author proposed the technique for greatest probability for arrangement alongside effective starter outputs. Fragmented remote detecting photo into a landslide just as non-landslide regions conveyed through related to the twofold limit approach under grouping along the histogram-based totally thresholding approach. They additionally utilized the sifting method for prominent masses depending on the shape & geomorphologic elements allow dispensing alongside fake regions.

In [7], the author offered a methodology considering landslide highlights identification under crust masses making use of the remote detecting technique. The author presented the GIS examination in which they registered rudimentary factors of pixels & entire snap shots taking after extraordinary landslides typologies moreover Kolmogorov-Smirnov coefficient for deciding relationship with developments & images. The surface channels meant to supply greater prominent characteristics under landslides areas permit differentiation among different area uses & landslides.

In [8], assessment for distant detecting data application considering landslide checking added alongside their methods & problems. At end author concluded as landslide expectation stays difficult & testing also along ground observation strategies.

In [9], they provided the smart solution for perceiving seismic tremor set off landslides spatial movement through examination later occasion nature contrasts. They applied before & after landslide remote sensing photos received through remote detecting over a comparable report area. They registered the radiometric standardization & mathematical adjustment for delivering each image vegetation record. At long last the direction of image difference calculation applied to recognize the change & produce the landslide map.

In [10], further method of landslide area diversity based on image processing is introduced. The authors created & assessed techniques for exchange identity over the automatic images of Tessina landslide underneath Italy. The different thresholding strategies are considered simply as the numerous channels used to put away the last annoying clutter from the thresholding resolution.

In [11], author introduced far off detecting primarily based technique for the considerable landslides inventory & image translation, examination of determinant factors, traits sound system plotting, & surface research-based car landslide discovery.

From above evaluations it is visible that the vast majority of the strategies depend on AI & thresholding method for the exchange discovery & landslide prediction. Greater part of overdue works zeroed below on semi-automatic approach alongside the limited association of datasets which drives the problems of adaptability simply because the lower & better pixel goals. Another charming problem is that no new work is reported over the Indian landslide areas, as a significant majority of the work comes from China. The basic levels of remote sensing imagery are techniques for preprocessing, segmentation, highlight extraction/selection, and landslide location, either thresholding or AI [11-21].

III. Landslide Prediction Approaches

The landslide prediction approaches focus on the support vector machine and artificial neural network machine learning algorithms. The detailed explanation is as follows.

A. SVM

SVM is a valuable method for information categories. Although it's taken into consideration that Neural Systems are less entangled to use than this, but all so often unsatisfactory results are received [22]. A category venture typically entails with training and checking out records which consist of a few facts' instances. Each example within the preparing set passes on one target esteems and different properties [23-24]. Arrangement in SVM is an instance of Managed getting to know. Regarded labels assist indicate whether the gadget is appearing appropriately or now not. This measurement focuses to an ideal response, approving the exactness of the contraption, or be utilized to help the machine figure out how to act effectively. A stage in SVM characterization incorporates ID as which are in detail identified with the perceived training. This is alluded to as feature decision or feature extraction. Capacity choice and SVM grouping together have utilization regardless of whether the forecast of obscure examples isn't basic. They might be utilized to recognize key units which may be worried about anything procedures recognize the exercises. The following image shows the landslide area using the SVM technique.

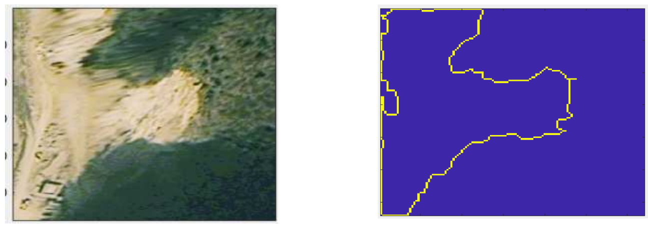

Fig. 1. SVM Technique

B. ANN

In system learning and intellectual science, artificial neural networks (ANNs) are a group of styles invigorated by the method for natural neural networks which can be utilized to evaluate or inexact abilities that can rely on a substantial assortment of information sources and are normally obscure. Counterfeit neural systems are ordinarily given as structures of interconnected "neurons" that substitute messages among one another. The associations have numeric loads that can be turned on a very basic level subject to acknowledge, making neural nets flexible to information sources and fit for learning. For instance, a neural system for penmanship ID is portrayed by methods for an arrangement of entering neurons which can be actuated with the guide of the pixels of an information photo [26-28]. In the wake of being weighted and changed through a trademark (chosen by utilizing the network's creator), the activation of those neurons is then passed explicitly to distinct kind of neurons. This technique is rehashed till at last, the output neuron that figures out which individual looked at is actuated. Like different gadgets gaining knowledge of strategies – frameworks that exploration from information – neural systems have been used to clear up a gigantic style of undertakings, like laptop vision and speech recognition, which can be hard to resolve the usage of normal rule-based programming. A synthetic neural system is an interconnected collecting of centers, figuring out with the in all cases system of neurons in a cerebrum. Artificial neural community sorts vary from people with the most effective one or layers of only one route logic, to complex multi–enter numerous directional comments circles and layers [29-30]. Overall, these structures utilize algorithms in their programming to choose to oversee and business venture of their capacities. Most extreme structures utilize "weights" to exchange the parameters of the throughput and the distinctive relationship with the neurons. Fake neural systems might act naturally adequate and considered by the method for entering from outside "teachers". The following image shows the landslide area using the ANN technique.

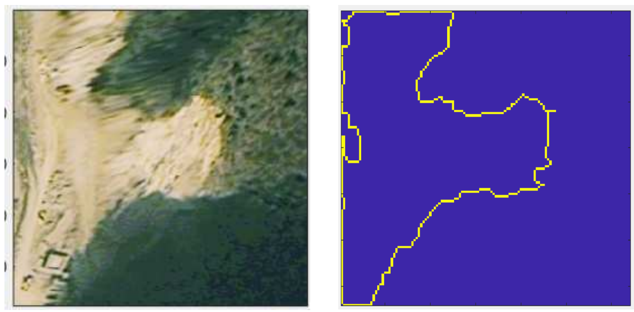

Fig. 2. ANN Technique

IV. Methodology

The following methodology is the proposed real-time landslide prediction and severity analysis which comprises of data collection, pre-processing, feature extraction, prediction and severity analysis as mentioned in Fig.3.

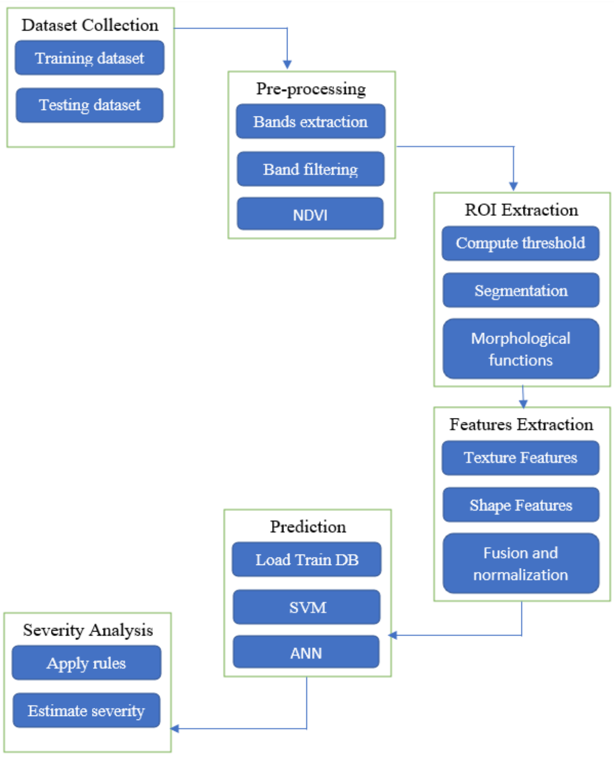

Fig. 3. Proposed real-time landslide prediction and severity analysis

A. Preprocessing

After the collection of satellite images of both pre and post landslide events, it is desirable to pre-process all the collected images. Before performing the image processing and analysis, it is required to first geometrically correct the remote sensing image by removing the vegetation, water, and built-up areas from the original images. Red and NIR Band extraction along with band filtering and NDVI computation is done for the selected image as shown in Fig.4.

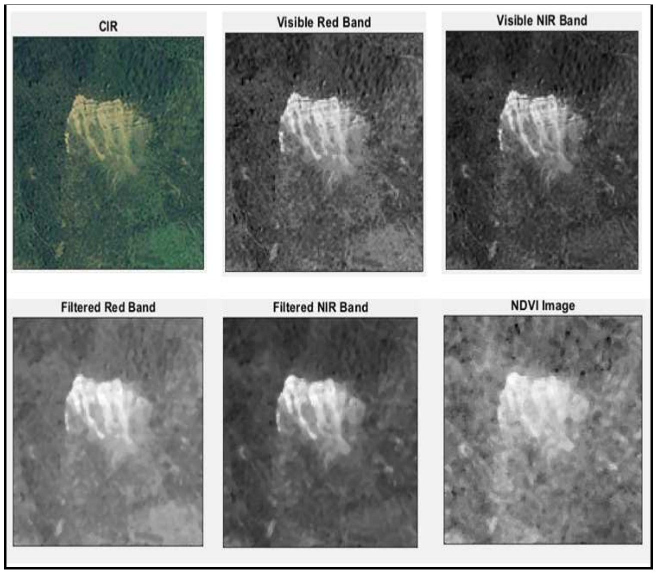

Fig. 4. Outcomes for the pre-processing of RS image1

B. Feature Extraction

After the pre-processing part, we perform the simple but effective segmentation approach based on dynamic thresholding and morphological operations which estimate the core areas of landslide reasons. From the segmentation image, based on hybrid feature extraction technique the change areas in pre and post landslide images can be identified efficiently. The texture and shape features combined together to make 28 features. Hybrid features includes texture and temporal features. The texture features are obtained using Gray Level Co-occurrence Matrix (GLCM) which includes contrast, energy, homogeneity, correlation and statistical features such as mean, standard deviation, variance, and entropy. To estimate the temporal variations in the input ROI image, we extracted the shape features. The geometric moment invariance is a suitable technique to represent the shape characteristics of the ROI image.

C. Prediction

After the features extraction and training of labeled images of before and after landslide images of selected areas, the classification performed by using the Artificial Neural Network (ANN) and Support Vector Machine (SVM) [31]. For the prediction of the possibility of landslides from the test RS sample, we have trained two different classifiers ANN and SVM. The ANN classifier trained the training dataset with hidden layers for hybrid features extraction is set to 7. We used the feed forward neural network for the training of 70 % of the data along with Levenberg-Marquardt backpropagation algorithm. Whereas, 30 % of test samples were used to perform the prediction. The design of SVM is done using the 10-fold cross-validation mechanism. SVM training, like ANN, is done on 70% of the training dataset while predictions are done on 30% of the test samples. The total data collection including landslide and non-landslide images is 2700 images [32]. The test image features passed to classifier in order to predict possibility landslide. If the average of the feature is less than 0.25 then the there is a lower possibility of the landslide. If the average of the feature is between 0.25 to 0.50 then the there is a medium possibility of the landslide. If average of the feature is greater than 0.50 then the there is a severe possibility of the landslide. The possibility of landslide in terms of three possible ranges is mentioned in following table.

TABLE I. POSSIBILITY OF LANDSLIDE

Feature Mean Value	Possibility of Landslide
Less than 0.25	Lower
0.25-0.50	Medium
Greater than 0.50	Severe

V. RESULT

Comparative analysis taking in to consideration Accuracy, F1 score, Precision, Recall for SVM and ANN is shown in Fig.5. From the comparative analysis it is interpreted that ANN performs better as compared to the SVM technique.

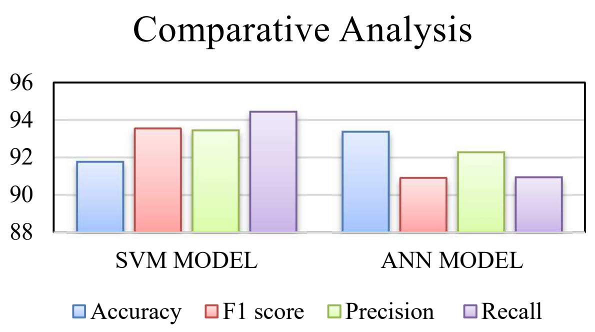

Fig.5. SVM & ANN Comparative Analysis

VI. CONCLUSION

It is feasible to create maps of probably events of future landslides using GIS and remote sensing. Landslide prediction and mapping can provide important data for crisis loss reduction, and it provide help in the development of sustainable land use planning. The mentioned comparative analysis between SVM and ANN gives us idea about selecting the appropriate soft computing technique which can be used for landslide prediction. As per the results ANN machine algorithm provides overall better accuracy of 93.38% as compared to SVM 91.78% regarding landslide prediction. It helps us save property damage, harm and loss of lives that adversely affect various natural resources. Landslide prediction helps to protect our country's water supplies, fisheries, sewage systems, forests, dams and roads.

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